

1 Model

For a factor return $f \in \mathbb{R}^q$ and specific return $\epsilon \in \mathbb{R}^p$, the excess security return,

$$(1) \quad y = Bf + \epsilon.$$

with f and ϵ uncorrelated. $\text{var}(y) = B\text{var}(f)B^\top + \text{var}(\epsilon)$ with $\text{var}(\epsilon)$ diagonal.

$$(2) \quad \text{var}(f) = \begin{pmatrix} 256 & 17 & -9 & 51 & 19 & 6 & 1 \\ 17 & 16 & 0 & 0 & 0 & 0 & 0 \\ -9 & 0 & 4 & 0 & 0 & 0 & 0 \\ 51 & 0 & 0 & 400 & 0 & 0 & 0 \\ 19 & 0 & 0 & 0 & 225 & 0 & 0 \\ 6 & 0 & 0 & 0 & 0 & 100 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 25 \end{pmatrix}$$

$$(3) \quad \text{corr}(f) = \begin{pmatrix} 1.00 & 0.28 & -0.30 & 0.16 & 0.08 & 0.04 & 0.02 \\ 0.28 & 1.00 & -0.11 & 0.00 & 0.00 & 0.00 & 0.00 \\ -0.30 & -0.11 & 1.00 & 0.00 & 0.00 & 0.00 & 0.00 \\ 0.16 & 0.00 & 0.00 & 1.00 & 0.00 & 0.00 & 0.00 \\ 0.08 & 0.00 & 0.00 & 0.00 & 1.00 & 0.00 & 0.00 \\ 0.04 & 0.00 & 0.00 & 0.00 & 0.00 & 1.00 & 0.00 \\ 0.02 & 0.00 & 0.00 & 0.00 & 0.00 & 0.00 & 1.00 \end{pmatrix}$$

Factor volatilities $\sigma_f = \sqrt{\text{diag}(\text{var}(f))}$ are given by,

$$(4) \quad \sigma_f = (16.0 \quad 4.0 \quad 2.0 \quad 20.0 \quad 15.0 \quad 10.0 \quad 5.0)$$

with expectations,

$$(5) \quad E(f) = (6.00 \quad 3.48 \quad 4.44 \quad 0.00 \quad 0.00 \quad 0.00 \quad 0.00)$$

Population means $\alpha = E(y) = BE(f) + E(\epsilon)$ have the averages,

$$p^{-1} \sum_{i=1}^p \alpha_i = 6.37 \quad p^{-1} \sum_{i=1}^p (BE(f))_i = 6.1 \quad p^{-1} \sum_{i=1}^p E(\epsilon_i) = 0.26$$

Population volatilities $\sigma_X = \sqrt{\text{diag}(\text{var}(X))}$ are given by,

$$p^{-1} \sum_{i=1}^p (\sigma_y)_i = 41.29 \quad p^{-1} \sum_{i=1}^p (\sigma_{Bf})_i = 21.26 \quad p^{-1} \sum_{i=1}^p (\sigma_\epsilon)_i = 34.82.$$

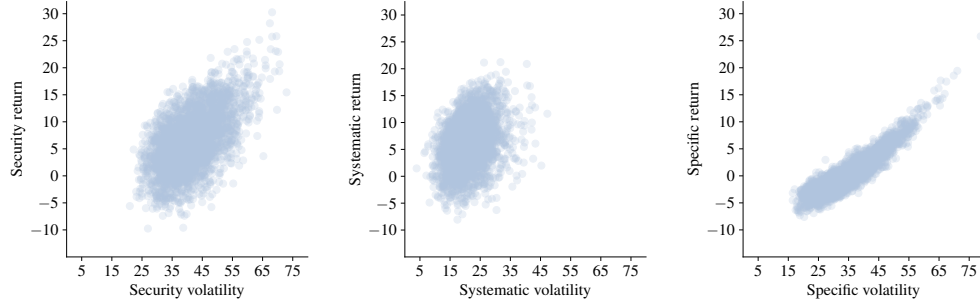


Figure 1: Scatter plots of α , $B E(f)$, $E(\epsilon)$ versus the corresponding volatilities.

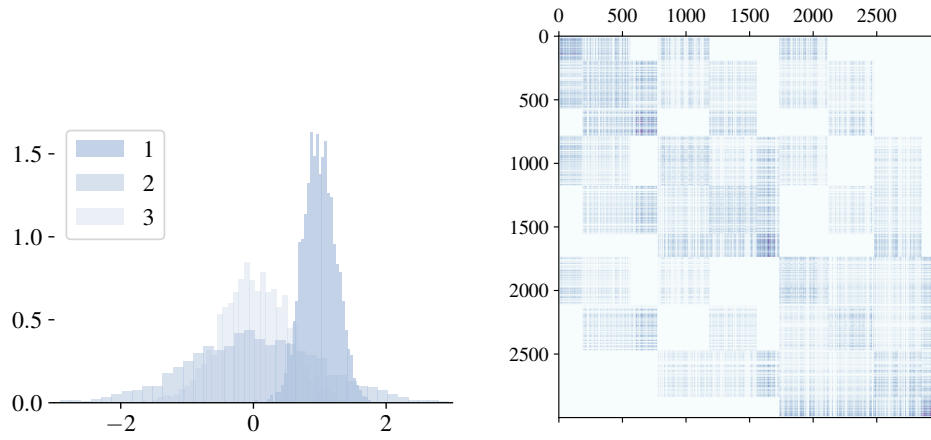


Figure 2: Left: Histograms of dense factor exposures in $\mathcal{E}$. Right: Reordered sparse factor component $\mathcal{S}$ of $\mathcal{E} \mathcal{E}^\top = \text{“dense”} + \mathcal{S}$

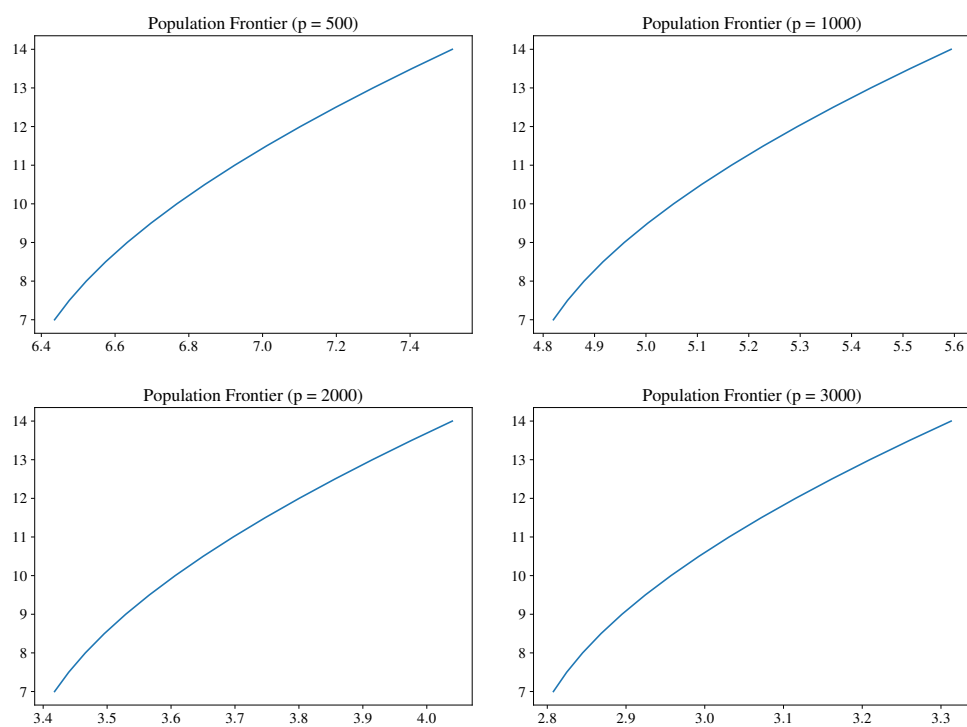


Figure 3: Effient (population) frontiers for each p .

2 Results

For $\hat{\Sigma}$ and estimate of the population covariance Σ , we solve

$$(6) \quad \begin{aligned} \min_{x \in \mathbb{R}^p} \quad & \frac{1}{2} \langle x, \hat{\Sigma} x \rangle \\ \langle m, w \rangle \geq & \mu, \\ \langle e, w \rangle = & 1, \end{aligned}$$

where the constraint is nonbinding in the (rare) cases when the minimum variance portfolio return exceeds μ . The constraint m is an estimate of $\alpha = E(y)$.

Let $\hat{w}$ be the solution of (26). The optimal portfolio w solves (26) with (Σ, α) replacing $(\hat{\Sigma}, m)$. We work with the following metrics.

$$(7) \quad \mathcal{V} = \sqrt{\langle \hat{w}, \Sigma \hat{w} \rangle} \quad (\text{True Volatility (TV)})$$

$$(8) \quad \hat{\sigma} = \sqrt{\langle \hat{w}, \hat{\Sigma} \hat{w} \rangle} \quad (\text{Estimated Volatility (EV)})$$

$$(9) \quad \sigma = \sqrt{\langle w, \Sigma w \rangle} \quad (\text{Optimal Volatility})$$

$$(10) \quad \Pi = \frac{\hat{\sigma}}{\mathcal{V}} \quad (\text{Precision})$$

$$(11) \quad \mathcal{R} = \frac{\langle \hat{w}, \alpha \rangle - r_f}{\mathcal{V}} \quad (\text{True Sharpe Ratio (TSR)})$$

$$(12) \quad \hat{\rho} = \frac{\langle \hat{w}, m \rangle - r_f}{\hat{\sigma}} \quad (\text{Estimated Sharpe Ratio (ESR)})$$

$$(13) \quad \rho = \frac{\langle w, \alpha \rangle - r_f}{\sigma} \quad (\text{Optimal Sharpe Ratio})$$

where r_f is the risk free rate.

The portfolio $\hat{w}$ decomposes into a minimum variance (MV) portfolio $\hat{w}_{\text{MV}}$ and the characteristic portfolio (CP) portfolio $\hat{w}_{\text{CP}}$, i.e.,

$$\hat{w} = \hat{w}_{\text{MV}} + \hat{w}_{\text{CP}}$$

This allows for a return to $\hat{w}$ to be decomposed.

- (14) $\langle \hat{w}, m \rangle$ (Estimated Return (ER))
- (15) $\langle \hat{w}, \alpha \rangle$ (True Return (TR))
- (16) $\langle \hat{w}_{\text{MV}}, m \rangle$ (Estimated MV Return (MVER))
- (17) $\langle \hat{w}_{\text{CP}}, m \rangle$ (Estimated CP Return (CPER))
- (18) $\langle \hat{w}_{\text{MV}}, \alpha \rangle$ (True MV Return (MVER))
- (19) $\langle \hat{w}_{\text{CP}}, \alpha \rangle$ (True CP Return (CPER))

p	OPT	PCA TV	PCA EV	$\mathcal{P}$	JSM TV	JSM EV	$\mathcal{P}$
500	6.57	9.1	6.03	0.66	8.08	8.35	1.03
1000	4.92	8.15	4.48	0.55	6.35	6.55	1.03
2000	3.5	7.26	3.17	0.44	4.71	4.71	1.0
3000	2.87	7.02	2.61	0.37	4.0	3.97	0.99

Table 1: Portfolio volatility statistics ($n = 125, \mu = 8.5$, 10 simulations).

p	OPT	PCA TSR	PCA ESR	$\mathcal{D}$	JSM TSR	JSM ESR	$\mathcal{D}$
500	1.29	0.72	1.7	2.36	0.85	1.21	1.43
1000	1.73	0.81	2.34	2.88	1.1	1.58	1.43
2000	2.43	0.92	3.32	3.61	1.52	2.22	1.46
3000	2.96	0.96	4.06	4.25	1.82	2.66	1.46

Table 2: True/Estimated Sharp Ratio (SR) statistics ($n = 125, \mu = 8.5$, 10 simulations).

p	OPT	PCA TR	PCA ER	$\mathcal{D}$	JSM TR	JSM ER	$\mathcal{D}$
500	8.5	6.54	8.5	1.3	6.85	8.5	1.24
1000	8.5	6.6	8.5	1.29	7.01	8.5	1.22
2000	8.5	6.68	8.5	1.27	7.18	8.5	1.19
3000	8.5	6.69	8.5	1.27	7.28	8.5	1.17

Table 3: True/estimated return statistics – True Return (TR) and Estimated Return (ER) – ($n = 125, \mu = 8.5$, 10 simulations).

p	PCA MV TR	PCA CP TR	PCA TR	PCA MV ER	PCA CP ER	PCA ER
500	3.05	3.49	6.54	3.34	5.16	8.5
1000	3.07	3.53	6.6	3.31	5.19	8.5
2000	3.33	3.34	6.68	3.55	4.95	8.5
3000	3.41	3.29	6.69	3.65	4.85	8.5

Table 4: PCA True/Esimated Return decompositions – Minimum Variance (MV) and Characteristic Portfolio (CP) – ($n = 125, \mu = 8.5$, 10 simulations).

p	JSM MV TR	JSM CP TR	JSM TR	JSM MV ER	JSM CP ER	JSM ER
500	3.46	3.39	6.85	4.4	4.1	8.5
1000	3.87	3.14	7.01	4.76	3.74	8.5
2000	4.96	2.22	7.18	5.81	2.69	8.5
3000	5.79	1.49	7.28	6.69	1.81	8.5

Table 5: JSM True/Esimated Return decompositions – Minimum Variance (MV) and Characteristic Portfolio (CP) – ($n = 125, \mu = 8.5$, 10 simulations).

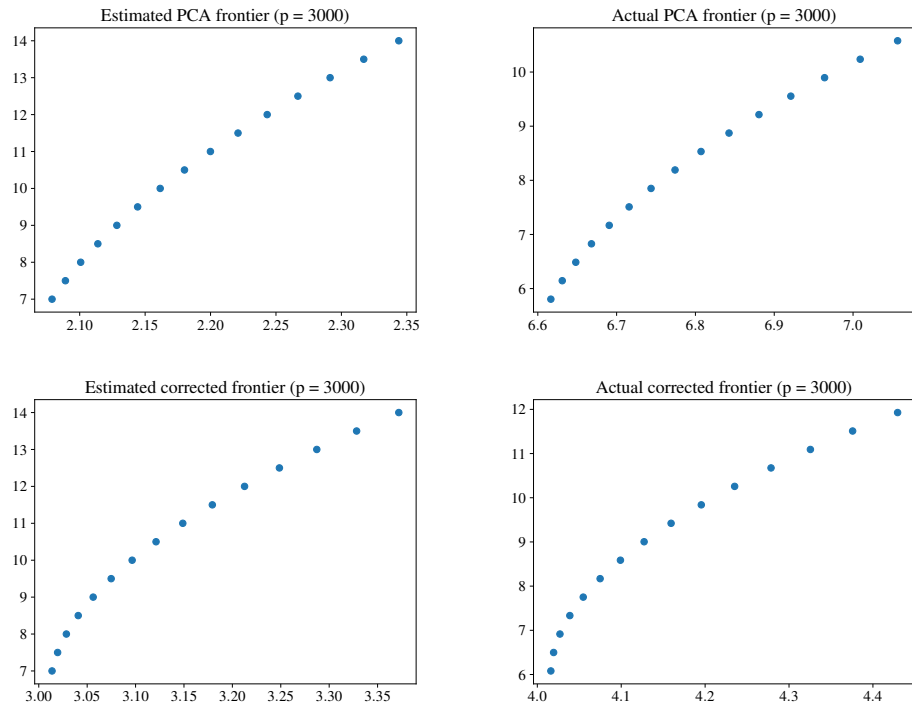


Figure 4: The actual (using Σ and α) and estimated (using $\hat{\Sigma}$ and m) frontiers for the simulation set in Table 1.

3 Recipes

1. Starting with $y = Bf + \epsilon$ we first write our model as

$$y = \alpha + Bx + u$$

where $x = (f - E(f))$ and $u = \epsilon - E(\epsilon)$ so that x, u have zero mean and,

$$\alpha = BE(f) + E(\epsilon) \quad (\text{and } E(\epsilon) \perp BE(f).)$$

2. From the model above we obtain the $p \times n$ data (for $\mathbb{1} = (1, \dots, 1) \in \mathbb{R}^n$),

$$(20) \quad Y = B\mathcal{F}^\top + \mathcal{E} = \alpha\mathbb{1}_n^\top + B\mathcal{X}^\top + \mathcal{U}.$$

3. The centered version of the data which we computed by

$$(21) \quad \bar{Y} = Y(I_n - \mathbb{1}\mathbb{1}^\top/n) = B\bar{\mathcal{X}}^\top + \bar{\mathcal{U}}.$$

where $\bar{\mathcal{X}} = \mathcal{X}(I_n - \mathbb{1}\mathbb{1}^\top/n)$ where $\bar{\mathcal{U}} = \mathcal{U}(I_n - \mathbb{1}\mathbb{1}^\top/n)$.

- Letting $\bar{x} = \mathcal{X}^\top \mathbb{1}/n$ and $\bar{u} = \mathcal{U}^\top \mathbb{1}/n$ we see that for each column x and z of $\mathcal{X}$ and $\mathcal{U}$ are of the form $x - \bar{x}$ and $u - \bar{u}$ in the matrices $\bar{\mathcal{X}}$ and $\bar{\mathcal{U}}$.
- $E(x - \bar{x})$ and $E(u - \bar{u})$ are not zero in general! (unless we have stationary). We assume stationarity for u (i.e. $E(u - \bar{u}) = 0$) but not for $x - \bar{x}$.
- We also have two sample covariance matrices.

$$S = YY^\top/n \text{ (uncentered)} \quad \bar{S} = \bar{Y}\bar{Y}^\top/n \text{ (centered)}$$

4. The mean vector for the data (that is subtracted during centering) is,

$$(22) \quad \bar{y} = \frac{Y\mathbb{1}}{n} = \alpha + B\bar{x} + \bar{u}$$

and our goal is to estimate $\alpha \in \mathbb{R}^p$. Our PCA model will use the constraint,

$$m = \bar{y}.$$

5. To design the JS estimator we follow the standard playbook. We take

$$\bar{y}(c) = c\bar{y} + (1 - c)g$$

for a shrinkage target g . For the grand mean target, $g = \langle \bar{y}, e \rangle e / p$,

6. Our portfolio will hedge out $B\bar{x}$ asymptotically, so we use JS to correct for $\bar{u}$. Standard calculations that minimize $|\bar{y}(c) - \alpha + B\bar{x}|^2$ give the optimal c as

$$(23) \quad c_* = \frac{\langle \bar{y} - g, \alpha - B\bar{x} - g \rangle}{|\bar{y} - g|^2}$$

7. Using the LLN behaviour of $\bar{u}$ we have the following asymptotics,

$$(24) \quad \langle \bar{u}, e \rangle / p, \langle \bar{u}, \alpha \rangle / p, \langle B\bar{x}, \bar{u} \rangle / p \rightarrow 0$$

8. From the above obtain the asymptotics,

$$(25) \quad c_* \sim 1 - \frac{|\bar{u}|^2}{|\bar{y} - g|^2}$$

The last term (i.e. $|\bar{u}|^2$) is easy to estimate with eigenvalues (see v^2 below).

$$m_{JS} = \left(1 - \frac{v^2}{|\bar{y} - g|^2}\right)m + \left(\frac{v^2}{|\bar{y} - g|^2}\right)g \quad (g = \langle \bar{y}, e \rangle e / p)$$

9. Our MJS estimator below will use m_{JS} in the Markowitz program.
10. Write $\bar{S} = HH^\top + G$ be the spectral decomposition of the (centered) sample covariance matrix and let n_+ denotes the number of its nonzero eigenvalues,

$$j_{1,p}^2 \geq j_{2,p}^2, \dots, j_{n_+,p}^2 > 0.$$

(If Y is full rank $n_+ = n - 1$). Above H is $p \times q$ matrix of q (spiked) eigenvectors,

$$H^\top H = \mathcal{S}^2$$

where $\mathcal{S}^2$ is the $q \times q$ matrix of spiked eigenvalues.

11. We write the constraint of the Markowitz program (26) in matrix form as

$$(26) \quad A = \begin{pmatrix} m_{JS} & e \end{pmatrix} \in \mathbb{R}^{p \times 2}$$

12. The average noise in the sample covariance is defined as

$$v^2 = \left(\frac{1 + n/p}{n_+ - q + n/p} \right) \sum_{j>q} \mathfrak{J}_{j,p}^2 \approx \frac{\sum_{j>q} \mathfrak{J}_{j,p}^2}{n_+ - q} \quad (n_+ > q)$$

13. We now arrive at our Markowitz-James-Stein estimator. Let $A^+ = (A^\top A)^{-1} A^\top$.

$$\begin{aligned} H^{\text{JSM}} &= HC + M(I - C) \\ M &= AA^+ H \\ N &= \mathfrak{S}^2 - H^\top AA^+ H \\ C &= (I - N^{-1} v^2) \end{aligned}$$

14. Alternatively,

$$\begin{aligned} H^{\text{JSM}} &= HC + M(I_q - C) \\ M &= AA^+ H \\ N &= (H - M)^\top (H - M) \\ C &= I - v^2 N^{-1} \end{aligned}$$

15. And alternatively again, for $\mathcal{H} = H \mathfrak{S}^{-1}$ (orthonormal columns),¹

$$\begin{aligned} \mathcal{H}^{\text{JSM}} &= \mathcal{H} C + \mathcal{M}(I - C) \\ \mathcal{M} &= AA^+ \mathcal{H} = \mathcal{A} \mathcal{A}^\top \mathcal{H} \\ N &= I - \mathcal{H}^\top AA^+ \mathcal{H} = I - \mathcal{H}^\top \mathcal{A} \mathcal{A}^\top \mathcal{H} \\ C &= I - N^{-1} v^2 \mathfrak{S}^{-2} \end{aligned}$$

¹Brief sketch, We write $HC + M(I - C) = \mathcal{H} \mathfrak{S} C + \mathcal{M} \mathfrak{S}(I - C)$ and off we go with

$$\begin{aligned} \mathfrak{S} C &= \mathfrak{S} - v^2 \mathfrak{S} N^{-1} = (I - v^2 \mathfrak{S} N^{-1} \mathfrak{S}^{-1}) \mathfrak{S} = (I - v^2 (\mathfrak{S}^{-1} N \mathfrak{S}^{-1})^{-1} \mathfrak{S}^{-2}) \mathfrak{S} = (I - v^2 (N')^{-1} \mathfrak{S}^{-2}) \mathfrak{S} \\ C' &= I - (N')^{-1} v^2 \mathfrak{S}^{-2} \end{aligned}$$

$$N' = \mathfrak{S}^{-1} N \mathfrak{S}^{-1} = I - \mathfrak{S}^{-1} H^\top AA^+ H \mathfrak{S}^{-1} = I - \mathcal{H}^\top AA^+ \mathcal{H}$$

$$\mathfrak{S}(I - C) = \mathfrak{S} v^2 N^{-1} = v^2 (\mathfrak{S}^{-1} N \mathfrak{S}^{-1})^{-1} \mathfrak{S}^{-1} = (N')^{-1} v^2 \mathfrak{S}^{-2} \mathfrak{S} = (I - C') \mathfrak{S}$$

Therefore, $\mathcal{H}^{\text{JSM}} = H^{\text{JSM}} \mathfrak{S}$.